

Daehee Park

Assistant Professor
DGIST (Daegu Gyeongbuk Institute of Science and Technology)
Department of Electrical Engineering and Computer Science

+82-10-3290-2155
dhpark@dgist.ac.kr
Google Scholar
Personal Website
ISL Lab Website

Research Interests

- **Multimodal Models for Physical AI**: Learning from vision, language, and motion for embodied understanding and action.
- **World Models and Motion Forecasting**: Modeling 3D environments, human motion, and multi-agent interactions to predict future states.
- **Prediction and Planning for Autonomous Systems**: Integrating perception, prediction, and planning for autonomous driving and robotics.
- **Efficient and Adaptive AI**: Model compression, model merging, and test-time adaptation for reliable deployment in dynamic environments.

Work Experience

- **DGIST (Daegu Gyeongbuk Institute of Science and Technology)**
  Assistant Professor, Dept. of Electrical Engineering and Computer Science — *2025.03 – Present*
  Assistant Professor, Dept. of Interdisciplinary AI — *2025.03 – Present*
- **Qualcomm**
  Interim Engineering Intern, QCT Auto AD — *2024.04 – 2024.09*
- **Naver Labs**
  Research Intern, Autonomous Driving Team — *2021.06 – 2021.08*

Education

- **KAIST (Korea Advanced Institute of Science and Technology)**
  Ph.D. in Mechanical Engineering — *2020.03 – 2025.02*
  Advisor: Prof. Kuk-Jin Yoon
- **KAIST (Korea Advanced Institute of Science and Technology)**
  M.S. in Mechanical Engineering — *2018.03 – 2020.02*
- **KAIST (Korea Advanced Institute of Science and Technology)**
  B.S. in Mechanical Engineering — *2013.03 – 2018.02*
  Double Major in Business and Technology Management

Publications

* equal contribution, † corresponding author

- **[c17]** Learning Scene-Grounded Interaction Priors for Scene-Aware Human Motion Prediction
  Donghyeon Soon*, Jiil Yoon*, Hyeondong Kim, **Daehee Park**†
  Neural Information Processing Systems **(NeurIPS)**, 2026.
- **[c16]** SRA: Spatial Reasoning Adapter via Evolving Social Interaction Graphs for Trajectory Prediction
  Jaewoo Jeong*, Seonkyu Song*, Hyeonwoo Park, Jegyeong Cho, Yujin Bae, Giwon Lee, **Daehee Park**, Kuk-Jin Yoon†
  Neural Information Processing Systems **(NeurIPS)**, 2026.
- **[c15]** Unified Prediction and Planning via Conflict-Aware Disjoint Parameter Training
  Taewon Seo*, Seonae Jeon*, Giwon Lee*, Kuk-Jin Yoon†, and **Daehee Park**†
  European Conference on Computer Vision **(ECCV)**, 2026.
- **[j2]** CascadeOcc: Rethinking 3D Occupancy World Models with Cascaded VQ Representations
  Kyumin Hwang*, Wonhyeok Choi*, Jaeyeul Kim, Jihun Park, **Daehee Park**†, Sunghoon Im†
  IEEE Signal Processing Letters, 2026.

- **[c14]** EggHand: A Multimodal Foundation Model for Egocentric Hand Pose Forecasting
  Jaeyoung Choi*, Hyeondong Kim*, Yujin Kim, **Daehee Park**†
  IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)** *Findings*, 2026.

- **[c13]** BitTP: The Lightweight Trajectory Prediction Model with BitLLM for Edge-Devices
  Mincheol Kang*, HyunJin Lim*, Bomin Kang*, **Daehee Park**†
  IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)** *Findings*, 2026.

- **[c12]** Generative Active Learning for Long-tail Trajectory Prediction via Controllable Diffusion Model
  Daehee Park, Monu Surana, Pranav Desai, Ashish Mehta, Reuben M. V. John, Kuk-Jin Yoon†
  IEEE/CVF International Conference on Computer Vision **(ICCV)**, 2025.

- **[c11]** Interaction-Merged Motion Planning: Effectively Leveraging Diverse Motion Datasets for Robust Planning
  Giwon Lee*, Wooseong Jeong*, **Daehee Park**, Jaewoo Jeong, Kuk-Jin Yoon†
  IEEE/CVF International Conference on Computer Vision **(ICCV)**, 2025 [Highlight].

- **[c10]** Non-differentiable Reward Optimization for Diffusion-based Autonomous Motion Planning
  Giwon Lee*, **Daehee Park***, Jaewoo Jeong*, Kuk-Jin Yoon†
  IEEE/RSJ International Conference on Intelligent Robots and Systems **(IROS)**, 2025.

- **[p1]** Self-Supervised 3D Occupancy Prediction with Temporal Consistency
  Jae-Seok Jeong, Sung-Hoon Yoon, **Daehee Park**, Kuk-Jin Yoon†.
  arXiv preprint, 2025.

- **[c9]** Multi-modal Knowledge Distillation-based Human Trajectory Forecasting
  Jaewoo Jeong, SeoHee Lee, **Daehee Park**, Giwon Lee, Kuk-Jin Yoon†
  IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)**, 2025.

- **[c8]** Diffusion-Guided Weakly Supervised Semantic Segmentation
  Sung-Hoon Yoon, Hoyong Kwon*, Jaeseok Jeong*, **Daehee Park**, Kuk-Jin Yoon†
  European Conference on Computer Vision **(ECCV)**, 2024.

- **[c7]** T4P: Test-Time Training of Trajectory Prediction via Masked Autoencoder and Actor-specific Token Memory
  Daehee Park, Jae-Seok Jeong, Sung-Hoon Yoon, Jaewoo Jeong, Kuk-Jin Yoon†
  IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)**, 2024.

- **[c6]** Multi-agent Long-term 3D Human Pose Forecasting via Interaction-aware Trajectory Conditioning
  Jaewoo Jeong*, **Daehee Park***, Kuk-Jin Yoon†
  IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR)**, 2024 [Highlight].
  Winner, Best Paper Award at Qualcomm Innovation Fellowship Korea

- **[c5]** Improving Transferability for Cross-domain Trajectory Prediction via Neural Stochastic Differential Equation
  Daehee Park, Jaewoo Jeong, Kuk-Jin Yoon†
  AAAI Conference on Artificial Intelligence **(AAAI)**, 2024.

- **[c4]** Leveraging Future Relationship Reasoning for Vehicle Trajectory Prediction
  Daehee Park, Hobin Ryu, Yunseo Yang, Jegyeong Cho, Jiwon Kim, Kuk-Jin Yoon†
  International Conference on Learning Representations **(ICLR)**, 2023.
  1st Place, nuScenes Prediction Challenge

- **[c3]** BIPS: Bi-modal Indoor Panorama Synthesis via Residual Depth-Aided Adversarial Learning
  Changgyoon Oh*, Wonjune Cho*, Yujeong Chae*, **Daehee Park***, Lin Wang, Kuk-Jin Yoon†
  European Conference on Computer Vision **(ECCV)**, 2022.
  Grand Prize, Best Paper Award at Workshop on Image Processing and Image Understanding 2022

- **[c2]** Unlocking the Potential of Ordinary Classifier: Class-specific Adversarial Erasing Framework for Weakly Supervised Semantic Segmentation
  Hyeokjun Kweon*, Sung-Hoon Yoon*, Hyeonseong Kim, **Daehee Park**, Kuk-Jin Yoon†
  IEEE/CVF International Conference on Computer Vision **(ICCV)**, 2021.

- **[j1, c1]** Identifying Reflected Images from Object Detector in Indoor Environment Utilizing Depth Information
  Daehee Park, Yong-Hwa Park†
  IEEE Robotics and Automation Letters **(RA-L)** *and IEEE International Conference on Robotics and Automation* **(ICRA)**, 2021.

Patents

- 궤적 예측에서 거대언어모델 경량화를 위한 비트 양자화 기법
  Korean Patent, Application, 10-2026-0156265, 2026.08.20.
- 태스크 간 충돌을 완화하는 파라미터 분리 학습 기반 모델 병합 방법 및 장치
  Korean Patent, Provisional application, 10-2026-0133460, 2026.07.21.
- 궤적 예측에서 거대언어모델 경량화를 위한 비트 양자화 기법
  Korean Patent, 10-2026-0093495.
- 1인칭 시점 손 동작의 인지, 예측 및 제어를 위한 멀티모달 파운데이션 모델 프레임워크 및 최적화 방법
  Korean Patent, 10-2025-0209315.
- Electronic device and method of identifying false image of object attributable to reflection in indoor environment thereof
  United States Patent, Registered, US11282178B2.
- Dispositif électronique et procédé d'identification de fausse image ou de faux objet imputable à la réflexion dans un environnement intérieur correspondant
  European Patent, Registered, EP3772701.
- 전자 장치 및 그의 실내 환경에서 반사에 의한 객체 허상을 식별하기 위한 방법
  Korean Patent, Registered, KR10-2287478-0000.

Positions of Responsibility

- Planning Committee Member, 2027 Physical AI R&D Projects, IITP, 2026.09
- Program Committee Member, IPIU 2027
- Planning Committee Member, 2026 Culture, Sports and Tourism R&D Projects, KOCCA, 2025.09

Awards and Honors

- Winner, NVIDIA Academic Grant Program, NVIDIA *2025.12*
- Outstanding Doctoral Graduate Award, KAIST ME *2025.12*
- Winner, Qualcomm Innovation Fellowship Korea (QIFK) *2024.12*
- Research Scholarship, Samsung Advanced Institute of Technology *2022.09 – 2025.02*
- 1st Place, Best Paper Awards, IPIU *2022*
- Outstanding Achievement Award, KAIST ME *2016.09*

Academic Reviewing & Service

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- IEEE/CVF International Conference on Computer Vision (ICCV)
- European Conference on Computer Vision (ECCV)
- Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- AAAI Conference on Artificial Intelligence (AAAI)

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Intelligent Vehicles Symposium (IV)
- Winter Conference on Applications of Computer Vision (WACV)
- ACM International Conference on Multimedia (ACM MM)
- British Machine Vision Conference (BMVC)
- Pattern Recognition (PR)
- IEEE Transactions on Intelligent Vehicles (T-IV)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- Engineering Applications of Artificial Intelligence (EAAI)

Research Grants

- 숙련자 시연데이터 기반 조선소 작업 특화 양팔 모바일 매니퓰레이터 및 Physical AI 작업스킬 실증 플랫폼 개발
  (PI) 산업통상부 / 한국산업기술기획평가원 *2026.08 – 2030.12*
- 도시형 생성 지능을 위한 사회적 군중 시뮬레이션 기술 개발
  과학기술정보통신부 / 인공지능산업융합사업단 (H100 x 8ea) *2026.07 – 2027.01*
- 희소 궤적 월드 액션 모델을 활용한 양손 조작 행동 생성 기술 개발
  과학기술정보통신부 / 한국전자기술연구원 (H100 x 8ea) *2026.07 – 2026.11*
- 경량형 온디바이스 인식 모델 최적화 연구 개발
  (PI) 경상북도 / 한국전자통신연구원 *2026.05 – 2027.12*
- 한국형 Physical LLM 의 실시간성과 안정성을 위한 가변형 추론 프레임워크 개발
  한국정보통신진흥협회 / 정보통신산업진흥원 (B200 x 16ea) *2026.05 – 2026.06*
- 다중 휴머노이드의 자율 협업을 위한 비전-언어 모델 기반 적응형 장기 · 복합작업 계획 기술
  과학기술정보통신부 / 한국연구재단 *2026.04 – 2030.12*
- 아카이브 영상 콘텐츠 활용을 위한 AI Outpainting 및 콘텐츠 변환 기술 개발
  (PI) 문화체육관광부 / 한국콘텐츠진흥원 *2026.04 – 2028.12*
- 자기 진화형 Physical LLM: 불확실한 환경에서 행동 파운데이션 모델의 강건하고 효율적인 적응 기술 연구
  (PI) 과학기술정보통신부 / 한국연구재단 *2026.03 – 2030.02*
- NVIDIA Academic Grant
  (PI) NVIDIA (Pro6000 x 4ea) *2026.02 – 2026.08*
- 행동 파운데이션 모델의 동적 환경으로 확장을 통한 안전하고 범용적인 인간-로봇 협동 연구
  (PI) 과학기술정보통신부 / 한국연구재단 *2025.09 – 2026.08*
- 로봇및기계전자공학연구소
  교육부 / 한국연구재단 *2025.09 – 2034.08*
- Physical AI 분야의 거대 생성모델 기술 선도를 위한 LG AI STAR 인재양성 사업
  (PI) 과학기술정보통신부 / 정보통신기획평가원 *2025.07 – 2028.12*
- (InnoCORE) 초거대언어모델 혁신 연구단
  과학기술정보통신부 / 한국과학기술원 *2025.07 – 2029.12*
- (InnoCORE) 바이오 체화형 피지컬 AI 연구단
  과학기술정보통신부 / DGIST *2025.07 – 2029.12*
- 초거대 클라우드팜 AI GPU 가상자원
  경북IT융합 산업기술원 (H100 x 2ea) *2025.07 – 2025.12*

- 생성형 AI 기반 16bit HDR 솔루션 개발
  삼성전자-KAIST 산학과제 *2025.05 – 2027.04*

- AI스타펠로우십지원(대구경북과학기술원)
  과학기술정보통신부 / 정보통신기획평가원(IITP) *2025.04 – 2028.12*

Invited Talks

- SMIT Pathfinding Seminar, Samsung Electronics Semiconductor R&D Center *2026.07.31*
  Title: Advances in Deep Learning Model Adaptation

- IEIE Image Understanding Summer School, Sookmyung Women's University *2026.07.15*
  Title: Research and Paper Writing in Computer Vision and Machine Learning from a Non-CS Background

- Young Leaders Session (YLS), IEIE Summer Conference *2026.06.24*
  Title: Self-Evolving Physical AI for Robust Decision Making

- KETI, Pangyo *2026.03.20*
  Title: Motion Prediction and Its Application for Physical AI

- ICT 4.0 Special Lecture, Hanbat National University *2025.11.24*
  Title: Prediction and Planning for Physical AI in Robotics & Autonomous Driving

- New Faculty Lunchtime Seminar, DGIST *2025.10.28*
  Title: Decision Making in Physical AI

- Young Researcher Talk, AISP 2025, Kyungpook National University *2025.09.26*
  Title: Prediction and Planning for Physical AI in Robotics and Autonomous Driving

- ETRI, Daegu-Gyeongbuk Research Center *2025.09.17*
  Title: Decision-making AI under Uncertainty: A Prediction and Planning Perspective

- EECS & AI Colloquium, DGIST *2025.09.09*
  Title: Prediction and Planning for Physical AI in Robotics & Autonomous Driving

- IEIE Joint Summer School on Image Understanding and Image Processing *2025.07.18*
  Title: Toward Decision-making AI: A Prediction and Planning Perspective

- 8th Daegu AI Researchers' Forum Special Seminar, Keimyung University *2025.05.30*
  Title: Data-driven Decision Making in Autonomous Systems: A Prediction and Planning Perspective

- KSAE Spring Conference, Convergence SW & AI Session *2025.05.22*
  Title: T4P: Test-Time Training of Trajectory Prediction via Masked Autoencoder and Actor-specific Token Memory

Teaching

Introduction to Deep Learning, Fall 2026.
Advanced Deep Learning, Spring 2026.
Artificial Intelligence and Machine Learning Engineering, Spring 2026.
Artificial Intelligence Basics, Fall 2025.
Computer Architecture, Spring 2025.
Deep Learning, Fall 2026.